CI/CD Pipeline for Python Energy Management Application using Docker

Course Name: DevOps

Institution Name: Medi-caps University – Datagami Skill Based Course

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Academic Year: 2025-26

Problem Statement & Objectives

1. Problem Statement

Modern AI-based research systems face several challenges in deployment and scalability. Traditional approaches often lead to:

- Inconsistent environments between development and production
- Complex integration of AI models and APIs
- Manual deployment processes prone to errors
- Difficulty in scaling AI workloads efficiently
- Slow processing of large document-based queries

These challenges reduce system efficiency and reliability. Therefore, an automated, containerized, and scalable solution is required.

2. Project Objectives

The main objectives of the Energy Management Application are:

- To develop an AI-powered energy research system
- To implement Retrieval-Augmented Generation (RAG)
- To automate deployment using CI/CD pipelines
- To containerize the application using Docker
- To enable scalable deployment using Kubernetes
- To ensure fast and accurate research report generation

3. Scope of the Project

The project focuses on:

- Building an AI-based research platform
- Implementing multi-agent architecture
- Integrating document-based retrieval (RAG)
- Automating CI/CD pipelines
- Containerizing application using Docker
- Deploying using Kubernetes for scalability
- Supporting efficient data processing and report generation

Limitations:

- Requires knowledge of AI and DevOps concepts
- Initial setup complexity for beginners
- Dependence on external APIs (LLM services)
- High computational requirements for AI processing
- Debugging multi-agent systems can be complex

4. Proposed Solution

The proposed system is an AI-powered Energy Management Application integrated with DevOps practices.

The solution includes:

- Using **Agentic AI** for automated research tasks
- Implementing **RAG (Retrieval-Augmented Generation)** for document-based insights
- Building backend APIs using FastAPI
- Creating interactive UI using Streamlit
- Containerizing the system using Docker
- Automating workflows using CI/CD pipelines (GitHub Actions/Jenkins)
- Deploying on Kubernetes for scalability and reliability

This ensures a robust, scalable, and production-ready system.

A. Key features

- AI-powered research report generation
- Multi-agent system for task automation
- RAG-based document processing
- Docker-based containerization
- Automated CI/CD pipeline
- Scalable deployment using Kubernetes
- Fast and efficient query processing

B. Overall Architecture / Workflow

The system follows a structured DevOps workflow:

1. User submits query through Streamlit UI
2. Request is sent to FastAPI backend
3. AI agents process the query
4. Documents are retrieved using FAISS vector database
5. LLM generates structured report
6. CI/CD pipeline automates build and deployment
7. Docker containers are deployed via Kubernetes
8. User receives final output

C. Tools & Technologies Used

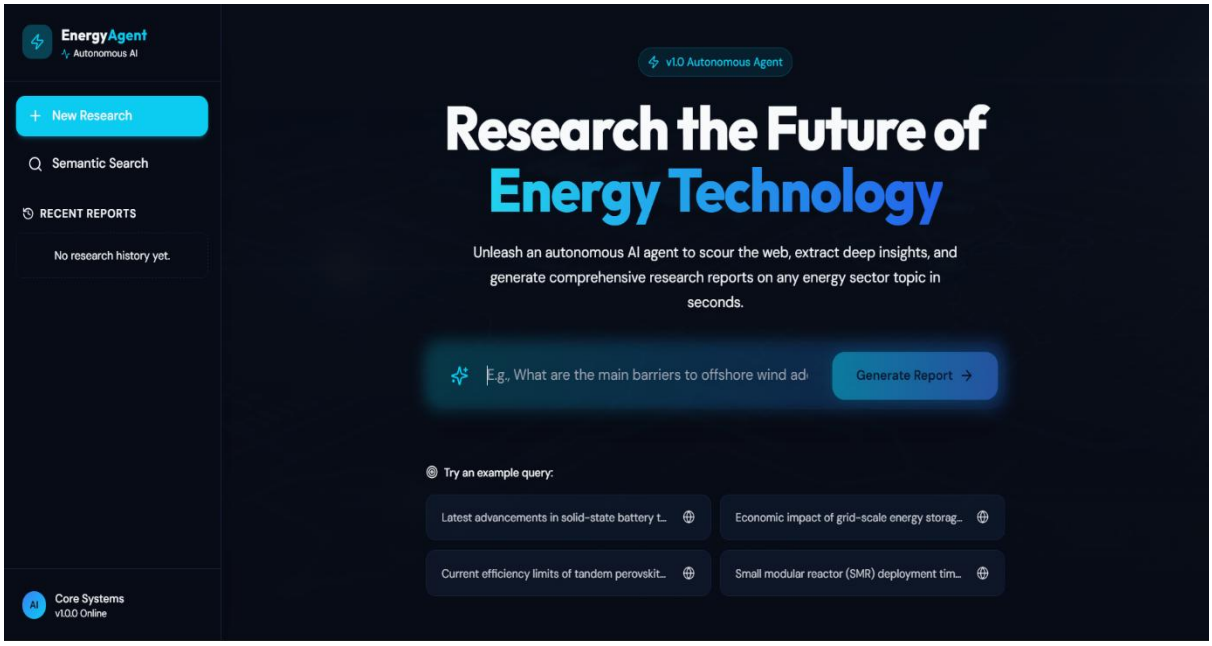
- **Programming Language:** Python
- **Frontend:** Streamlit
- **Backend:** FastAPI
- **AI Framework:** LangChain / CrewAI
- **Vector Database:** FAISS
- **LLM:** Groq (Mixtral Model)
- **Containerization:** Docker
- **CI/CD Tools:** GitHub Actions

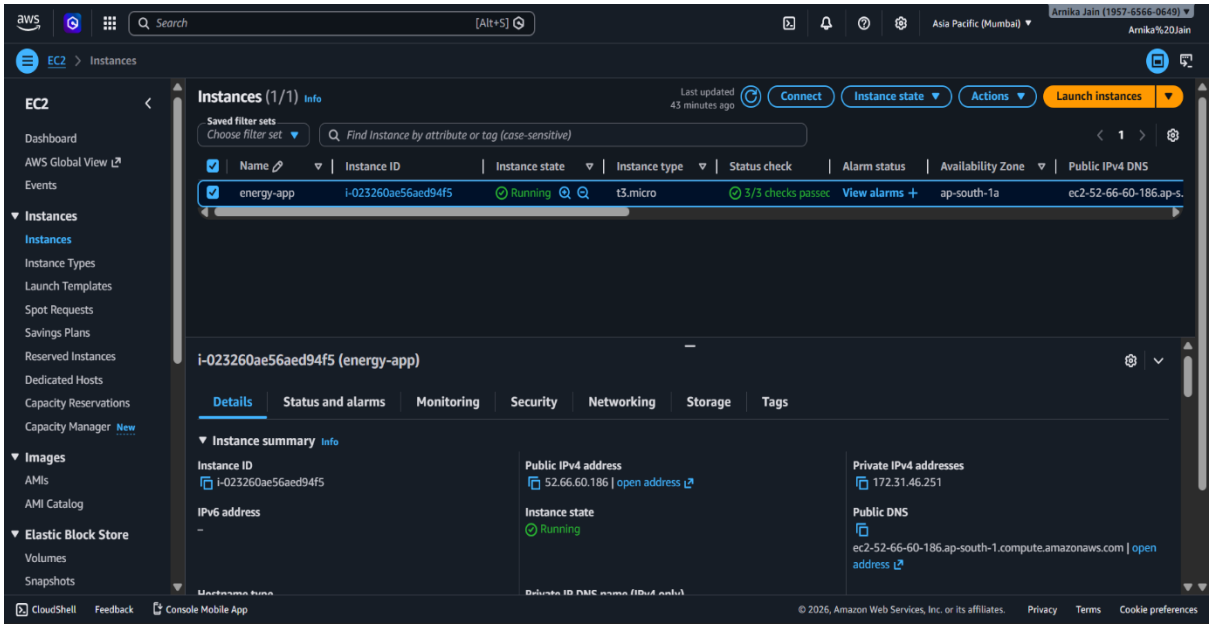
Results & Output

1. Screenshots / outputs

The project successfully demonstrates:

- Faster and automated deployment process
- Consistent environment across all stages
- Reduced manual configuration errors
- Improved scalability and system reliability


EnergyAgent
Autonomous AI
New Research
Semantic Search
RECENT REPORTS
No research history yet.
Core Systems
v1.0.0 Online
v1.0 Autonomous Agent
Research the Future of
Energy Technology
Unleash an autonomous AI agent to scour the web, extract deep insights, and generate comprehensive research reports on any energy sector topic in seconds.
E.g., What are the main barriers to offshore wind ad
Generate Report
Try an example query:
Latest advancements in solid-state battery t...
Economic impact of grid-scale energy storag...
Current efficiency limits of tandem perovskit...
Small modular reactor (SMR) deployment tim...


aws
Search
[Alt+S]
Asia Pacific (Mumbai)
Arnika Jain (1957-6566-0649)
Arnika%20Jain
EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Instances (1/1) Info
Last updated 43 minutes ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Find instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Alarm status
Availability Zone
Public IPv4 DNS
energy-app
i-023260ae56aed94f5
Running
t3.micro
3/3 checks passed
View alarms +
ap-south-1a
ec2-52-66-60-186.ap-s.
i-023260ae56aed94f5 (energy-app)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-023260ae56aed94f5
IPv6 address
-
Public IPv4 address
52.66.60.186 | open address
Instance state
Running
Private IPv4 addresses
172.31.46.251
Public DNS
ec2-52-66-60-186.ap-south-1.compute.amazonaws.com | open address
CloudShell
Feedback
Console Mobile App
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Cookie preferences

2. **Key outcomes**
   - Reduced deployment time
   - Improved system efficiency
   - Faster query processing
   - Enhanced scalability
   - Better automation and reliability

Conclusion

The Energy Management Application demonstrates the integration of **AI and DevOps** to build a modern intelligent system.

The system:

- Automates research processes
- Ensures consistent deployment using Docker
- Improves scalability using Kubernetes
- Enhances efficiency using CI/CD pipelines

This project provides practical exposure to real-world technologies like AI agents, containerization, and DevOps automation.

Future Scope & Enhancements

- Deployment on cloud platforms (AWS, Azure, GCP)
- Integration with monitoring tools (Prometheus, Grafana)
- Support for multiple LLM models
- Advanced UI/UX improvements
- Real-time analytics dashboard
- Enhanced security and authentication